

SplitFed Speech Emotion Classification

Comparative Experiment Tables

Summary tables

Table 1: Train/test split statistics for the E0 cross-corpus setup.

Dataset	Split	Samples	Actors	Anger	Non-anger	Anger share
CREMA-D	Train	5890	72	1006	4884	17.08%
CREMA-D	Test	1552	19	265	1287	17.07%
RAVDESS	Train	1140	19	152	988	13.33%
RAVDESS	Test	300	5	40	260	13.33%
SAVEE	Train	360	3	45	315	12.50%
SAVEE	Test	120	1	15	105	12.50%

Table 2: Aggregated dataset-level statistics for the E0 setup.

Dataset	Samples	Actors	Anger	Non-anger	Mean dur. (s)	Std. dur. (s)
CREMA-D	7442	91	1271	6171	2.5429	0.5060
RAVDESS	1440	24	192	1248	3.7007	0.3367
SAVEE	480	4	60	420	3.8395	1.0780

Table 3: Pairwise Maximum Mean Discrepancy (MMD) across datasets without normalization.

Dataset	CREMA-D	RAVDESS	SAVEE
CREMA-D	-	0.381637	0.389779
RAVDESS	0.381637	-	0.332834
SAVEE	0.389779	0.332834	-

Table 4: Pairwise Maximum Mean Discrepancy (MMD) across datasets after normalization.

Dataset	CREMA-D	RAVDESS	SAVEE
CREMA-D	-	0.009802	0.018052
RAVDESS	0.009802	-	0.017652
SAVEE	0.018052	0.017652	-

Table 5: E1 local-corpus evaluation at seed 42: anger F1 on each cross-corpus test set after 50 local epochs. Every model is trained only on the corpus named in its row.

Train model	CREMA-D test	RAVDESS test	SAVEE test
CREMA-D	0.6797	0.2222	0.6207
RAVDESS	0.4453	0.6364	0.4400
SAVEE	0.3224	0.2476	0.5217

Table 6: E2 centralized versus FedAvg quality at seed 42. Metrics are sample-weighted across the three corpus-specific test sets.

Method	Accuracy	Anger F1	Macro F1	Precision	Recall	PR-AUC
Centralized	0.8920	0.6864	0.8102	0.6655	0.7409	0.7772
FedAvg	0.8966	0.6598	0.7990	0.7101	0.6645	0.7757

Table 7: E3 protocol comparison at round 50 of the seed-42 rerun. Metrics are sample-weighted across the three corpus-specific test sets. The protocols retain their native workload and server-model ownership semantics.

Method	Accuracy	Anger F1	Macro F1	Precision	Recall	PR-AUC
MergeSFL	0.8763	0.6330	0.7786	0.6632	0.6845	0.7290
SFLv1	0.8834	0.6556	0.7923	0.6485	0.6914	0.7518
SFLour	0.8727	0.6498	0.7854	0.6259	0.7313	0.7691

Table 8: E4 SFLour workload-policy quality at round 50 of the seed-42 rerun. Metrics are sample-weighted across the three corpus-specific test sets.

Policy	Accuracy	Anger F1	Macro F1	Precision	Recall	PR-AUC
Full Local Epoch	0.8727	0.6498	0.7854	0.6259	0.7313	0.7691
Fixed Maximum Steps	0.8925	0.6601	0.7980	0.7040	0.6426	0.7292
Balanced Workload	0.8595	0.6265	0.7696	0.5723	0.7206	0.7113

Table 9: E4 SFLour workload-policy cost at round 50 of the seed-42 rerun. Batches/round are listed as CREMA-D/RAVDESS/SAVEE. The fixed budget of 143 is the RAVDESS (middle client) epoch length at batch size 8. CPU covers training roles and transport uses binary GiB. Per-role memory is reported separately in Tables 11 and 12.

Policy	Batches/round	Wall (s)	CPU user (s)	CPU sys (s)	Transport (GiB)	Messages
Full Local Epoch	737/143/45	780.25	1604.00	285.90	22.869	95150
Fixed Maximum Steps	143/143/143	289.91	812.77	119.00	10.772	45550
Balanced Workload	64/64/45	174.77	334.77	51.91	4.531	19950

Table 10: Runtime and communication cost for the E1–E4 seed-42. Wall time is the end-to-end experiment interval; CPU times are summed across training roles. E1 did not record an end-to-end experiment interval, so its wall time is unavailable. Transport uses binary GiB.

Method	Wall (s)	CPU user (s)	CPU sys (s)	Transport (GiB)	Messages
E1 Local	–	338.87	83.61	0.000	0
E2 Centralized	521.39	351.03	85.20	0.000	0
E2 FedAvg	561.27	2712.32	150.79	1.959	300
E3 MergeSFL	251.42	404.41	82.73	7.320	31520
E3 SFLv1	849.79	4089.91	336.52	22.869	95300
E3 SFLour	780.25	1604.00	285.90	22.869	95150
E4 Balanced	174.77	334.77	51.91	4.531	19950
E4 Fixed steps	289.91	812.77	119.00	10.772	45550

Table 11: Topology-aware owned tensor footprint in binary MiB. Max client is the requirement for one client node; client and SplitServer sums are logical footprints across isolated participants, not process RSS. A dash denotes a role absent from the selected topology.

Method	Client max	Clients sum	Central trainer	Split max	Split sum	FedServer	Logical total
E1 Local	125.58	196.38	–	–	–	–	196.38
E2 Centralized	–	–	192.89	–	–	–	192.89
E2 FedAvg	153.40	232.93	–	–	–	6.68	239.62
E3 MergeSFL	133.51	173.26	–	6.55	6.55	0.13	179.94
E3 SFLv1	133.77	174.05	–	19.63	58.88	0.13	233.07
E3 SFLour	133.77	174.05	–	19.63	19.63	0.13	193.81
E4 Balanced	133.77	174.05	–	19.63	19.63	0.13	193.81
E4 Fixed steps	133.77	174.05	–	19.63	19.63	0.13	193.81

Table 12: Maximum per-process RSS/CUDA allocated memory by runtime role, in binary MiB. Each cell is RSS/CUDA. Peaks from different processes are not summed; for SFLv1, the SplitServer cell is the maximum of its three isolated personalized model processes.

Method	Client	Central trainer	SplitServer	FedServer
E1 Local	2032.29/149.35	–	–	–
E2 Centralized	–	2061.59/149.35	–	–
E2 FedAvg	2066.53/149.35	–	–	806.36/2.25
E3 MergeSFL	1873.99/10.73	–	1555.88/177.22	766.96/0.14
E3 SFLv1	1845.88/11.00	–	1656.07/142.42	766.21/0.14
E3 SFLour	1848.27/11.00	–	1627.85/239.07	766.70/0.14
E4 Balanced	1832.95/11.00	–	1627.76/239.07	766.65/0.14
E4 Fixed steps	1862.80/11.00	–	1623.53/239.07	767.18/0.14

Table 13: Per-client E1 local-training resources under Resource Metrics v2 (seed 42, 50 epochs). Owned memory counts tensor storage attributable to one participant; RSS and CUDA columns are per-process peaks and are not summed. All memory values use binary MiB. Evaluation runs in separate processes and is excluded.

Local corpus	Epoch time sum (s)	Owned	Dataset	Process RSS	Phase RSS	CUDA alloc.	CUDA reserved
CREMA-D	341.52	125.58	105.56	2032.29	2033.63	147.32	216.00
RAVDESS	67.52	41.56	21.54	1906.09	1907.88	147.58	216.00
SAVEE	21.73	29.24	9.21	1897.51	1898.84	149.35	218.00
Logical sum / physical max	430.78	196.38	136.31	2032.29	2033.63	149.35	218.00

Table 14: E1 local cross-corpus anger F1 over seeds 42–47. Values are mean $\pm$ sample standard deviation across six runs. Each seed produces a new actor-disjoint split.

Train corpus	CREMA-D test	RAVDESS test	SAVEE test
CREMA-D	0.695 ± 0.015	0.040 ± 0.055	0.334 ± 0.253
RAVDESS	0.344 ± 0.146	0.651 ± 0.139	0.264 ± 0.146
SAVEE	0.312 ± 0.064	0.192 ± 0.098	0.488 ± 0.255

Table 15: E2 centralized and FedAvg anger F1 over seeds 42–47. Values are mean $\pm$ sample standard deviation across six actor-disjoint splits.

Method	CREMA-D	RAVDESS	SAVEE
Centralized	0.696 ± 0.032	0.649 ± 0.077	0.270 ± 0.246
FedAvg	0.727 ± 0.028	0.353 ± 0.146	0.338 ± 0.215

Table 16: E3 and E4 quality and runtime over seeds 42–47. Quality columns are macro-averaged across corpora except for weighted anger F1. Values are mean $\pm$ sample standard deviation across six runs. Runtime is wall-clock time in seconds.

Method	Accuracy	Anger F1	Macro F1	UAR	PR-AUC	Weighted anger F1	Time (s)
MergeSFL	0.820 ± 0.123	0.519 ± 0.080	0.692 ± 0.092	0.746 ± 0.027	0.662 ± 0.086	0.601 ± 0.050	262.2 ± 8.5
SFLv1	0.842 ± 0.083	0.532 ± 0.105	0.711 ± 0.081	0.740 ± 0.053	0.683 ± 0.068	0.643 ± 0.032	837.9 ± 19.3
SFLour full epoch	0.837 ± 0.093	0.535 ± 0.076	0.709 ± 0.062	0.732 ± 0.045	0.676 ± 0.066	0.640 ± 0.026	832.2 ± 15.9
SFLour balanced	0.808 ± 0.111	0.551 ± 0.063	0.701 ± 0.080	0.758 ± 0.046	0.644 ± 0.059	0.620 ± 0.021	187.2 ± 4.7
SFLour fixed steps	0.842 ± 0.086	0.560 ± 0.071	0.725 ± 0.063	0.752 ± 0.046	0.695 ± 0.054	0.636 ± 0.021	316.6 ± 17.0